CO1

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Course Name : Computer Networks

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Total Marks Obtained: 20

Faculty incharge: 20/7/26

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HOD :

File Transfer (150MB)

Given,

File size = 150MB = 150 × 1024 × 1024 = 157286400 bytes ≈ 150×10^6

Segment size = 1500 bytes

Data Link overhead / frame = 40 bytes / frame

Bandwidth = 50 Mbps.

Calculation:

No. of segments = 150,000,000 ÷ 1500
= 100000 segments.

No. of frames = 100000 frames

Total overhead = 100,000 × 40
= 4000 000 bytes (4MB)

Total data transmitted = 150 MB + 4MB
= 154 MB

Convert to bits : 154 × 8 = 1232 Mb

Transmission time = 1232 ÷ 50
= 24.64 seconds.

Explanation:

→ Transport Layer divides data into segments and ensures reliable delivery using acknowledgements and error recovery.

→ Data Link Layer creates frames, detects errors using CRC, and provides reliable node-to-node communication.

2. Sliding Window

Given, window size = 6
Total frames = 48

Calculation:

Windows required = 48 ÷ 6
= 8 windows

One frame lost in each window

Retransmissions = 8.

Explanation:

Flow control prevents sender overflow by matching transmission speed with receiver capacity, improving efficiency and reducing congestion.

3 IP Fragmentation

Given, Packet = 12,000 bytes
MTU = 1500 bytes
IP header = 20 bytes

Calculation

Payload per fragment = 1500 − 20
= 1480 bytes

Fragments required = Ceiling (12000 ÷ 1480)
= 9 fragments.

Header overhead = 9 × 20 = 180 bytes.

The Network Layer fragments packets when MTU is exceeded and reassembles them at the destination using identification, offset & flags.

liding Window

Given, Window size = 12

Total frames = 48

Windows required = 48 ÷ 12 = 4 windows

One frame lost in each window.

Retransmissions = 4

Flow control prevents sender overflow by matching transmission speed with receiver capacity, improving efficiency & reducing congestion.

5. Session Layer

Given, File = 600 MB

Checkpoint every 60MB

Failure after 390MB

Checkpoints before failure = 390 ÷ 60

= 6 checkpoints.

Last checkpoint = 360MB.

Data to retransmit = 390 - 360

= 30MB.

Session layer synchronization checkpoints allow transmission to resume from the last checkpoint instead of restarting the entire transfer.

6. Presentation Layer

Given, Original file = 900 MB

Compression = 40%

Encryption overhead = 5%

Compressed size = 900 × 60% = 540 MB

Encrypted size = 540 × 1.05 = ~~M~~ 567 MB

Reduction = 900 − 567 = 333 MB

Percentage reduction = (333 ÷ 900) × 100

= 37%

The presentation layer compressies data to reduce transmission time and encrypts it to provide confidentiality and security.

7. FTP Application

Given, Data = 3GB = 3000 MB

Request size = 3MB

Throughput = 120 Mbps

Requests = 3000 ÷ 3 = 1000 requests

Data in bits → 3GB = 24,000 Mb.

Transmission time = 24000 ÷ 120 = 200 seconds.

The application layer provides services such as FTP, file transfer management, authentication, & error reporting.

nd-to-End Delay

Processing delay

Application = 4ms

Network = 5ms

Transport = 3ms

Data Link = 2ms

Physical = 1ms

Total processing = 15ms

Total delay = 15+18+12

= 45ms

Each OSI layer performs specific processing before transmission. Propagation delay depends on distance, while transmission delay depends on bandwidth.

9. Protocol Overhead.

Given, Useful data = 80MB = 80.000.000 bytes

frame = 1600 bytes

Overhead = 80 bytes

Useful payload / frame = 1600 − 80 = 1520 bytes

Frames required = 80,000,000 ÷ 1520 = 52,632 frames

Total Overhead = 52632 × 80 ≈ 4·21 MB.

Efficiency = (1520 ÷ 1600) × 100 = 95%.

Protocol overhead increases transmitted data & slightly reduces network efficiency, but it enables addressing, error detection, & reliable communication.

10. Healthcare Network

Given, Original data = 240MB = 240,000,000 bytes
Compression = 30%
Segment = 1400 bytes

Compressed size = 240 × 70% = 168 MB

Segments = 168000000 ÷ 1400 = 120000 segments.

Protocol overhead = 120000 × 60 = 7200 000 bytes.

Final transmitted data = 168 + 7.2
= 175.2 MB

→ Presentation layer compresses & encrypts data.
→ Transport layer divides data into segments.
→ Data Link layer adds frame headers/ trailers & performs error detection.
→ Physical layer transmits the final bit stream over the communication medium.

20/4/25

Understanding Concepts	Explanation	Application	Organization	Accuracy	Clarity
25%	25%	20%	15%	10%	5%